# Semester Master Plan and Workflow SOP

Version: 0.1 | Prepared: October 7, 2026 | Status: implementation proposal

## Purpose and status

Connect syllabus milestones, assignments, weekly study blocks and released-grade tracking through canonical references. The companion dashboard is a local HTML prototype with sample data, in-memory state and no live service connection. It is not a second production application. Integrate its approved interactions into Semester's existing shell, store, authorization and domain services after repository reconciliation.

## Evidence basis

- The Main Semester design system (2) copy/ui_kits/ecosystem/Ecosystem.jsx: proposed environment boundaries, record/audit/outbox flow and authorized projections; explicitly not verified as built.
- The Main Semester design system (2) copy/docs/system/workflows/SEMESTER_WORKFLOW_ARCHITECTURE.md: versioned workflow definition and universal lifecycle; dated October 5, 2026.
  These are design evidence, not live implementation verification. No claims about live endpoints, schema, deployments or institutional approval are made here.

## Canonical master plan

Each course links to the student's authorized section and enrollment context. Each assignment links to its course, syllabus milestone, source version, due time and course AI policy. Each private study block links to an assignment but remains owned by the student. Each released result links to an assessment attempt and grading authority. The master view is a projection, not another source of truth.

Required fields:
Course: canonical ID, section ID, term, institution scope, enrollment relationship, source.
Milestone: canonical ID, course ID, title, source version, date, confirmation status.
Assignment: canonical ID, milestone ID, title, deadline/timezone, workflow state, source version, freshness, grading category, points possible.
Study block: personal scope ID, owner ID, assignment reference, start/end/timezone, status, revision.
Released result: assessment/attempt reference, released score, possible score, category, release time, source version, authority.
Projection: authorized viewer scope, source versions, updated time, delivery state.

A planned completion is not a submission. A self-marked completion is not an institutional receipt. A predicted score is not a released grade. Missing scores are unknown, not zero.

Overdue is derived from authoritative due time and confirmed submission state, not a universal assignment state.

### Owners and handoffs

Student: choose study blocks, edit private plans, review feedback and request support.
Faculty/authorized teaching team: publish milestones, confirm deadlines, define grading rules and release feedback within authority.
Advisor/tutor: receive only permitted referrals or explicitly shared context.
Platform engineering: canonical references, commands, delivery, projections and reconciliation.
Security/privacy: authorization, classifications, consent, retention and AI data scope.
Institution domain owner: authorize institutional writes and resolve source disputes.
Support: triage minimized cases; engineering receives sanitized telemetry.

### Start-of-term SOP

1. Establish active identity, institution membership and course relationships.
2. Import or reference permitted syllabus sources; record provenance and source version.
3. Extract candidate milestones, deadlines and grading rules as unconfirmed drafts.
4. Compare with authoritative course records. Resolve contradictions with faculty or the designated source owner; never let AI decide authority.
5. Confirm timezone, due-time conventions, category weights, dropped-score rules, late penalties and extra-credit handling.
6. Create assignment references once; Calendar, Action Center and Study consume those references.
7. Capture student availability, preferred study cadence and notification settings privately.
8. Propose study blocks before deadlines with travel, rest and contingency buffers chosen by the student.
9. Preview overlaps and uncertain deadlines. The student confirms private scheduling.
10. Record plan revision and source versions. Show unknowns and blockers in the master view.
    Exit gate: course relationships authorized, all extracted facts labeled, uncertain grading/deadline rules visible, no fabricated grade estimate.

### Weekly operating SOP

Monday or student-selected planning day:

1. Refresh course sources and released results. Display stale or unavailable data.
2. Review milestones and assignments for the next two weeks.
3. Compare estimated effort with private availability; estimates are student-editable, not facts.
4. Add or adjust blocks, check overlaps and end times, then confirm changes.

5. Route source disputes or institutional blockers to the correct office.
   During the week:
6. Open the assignment context from Action Center; use permitted materials and AI rules.
7. Record private progress. Submit through the authorized submission workflow separately.
8. Display a submission receipt only after authoritative confirmation; reconcile unknown outcomes.
   End of week:
9. Review released feedback and recalculated grade summaries.
10. Reschedule unfinished work, revise effort estimates and request help when chosen.
11. Archive completed private blocks without deleting institutional evidence.
    Exit gate: no orphaned blocks, visible source freshness, actual receipts distinguishable from personal completion, pending deliveries assigned an owner.

### Sample week: October 7–13, 2026

All entries are fictional examples, not the user's courses.
BIO 110 / Cell biology quiz: due October 9 at 17:00; target 2 hours; initial private block October 8 at 14:00 for 60 minutes.
ENG 210 / Argument essay: due October 12 at 23:59; target 4 hours; initial private block October 9 at 10:00 for 90 minutes.
MATH 150 / Problem set 4: due October 8 at 18:00; sample state Submitted; no released score.
BIO 110 / Lab report 2: released sample score 42/50.
MATH 150 / Problem set 3: released sample score 27/30.
ENG 210 / Reading response: due October 10 at 12:00; stale source; sample scheduling blocked until refresh.
The dashboard uses browser-local times only. Production must use institution/course IANA timezone plus UTC timestamps; DST behavior requires testing.

### Grade-tracking SOP

Point subtotal = sum(released earned points) / sum(released possible points), using only the applicable released attempts and course rules. Label this as a released-point subtotal, not a final grade. Do not average percentages when point values differ.
Weighted grade calculations require validated category rules. For each category, apply the institution's approved handling of attempts, drops, penalties and extra credit. Show which categories contain released evidence and the proportion of syllabus weight represented. Do not normalize partial evidence without labeling that choice.
Predictions and what-if results belong in a separate personal estimate, with assumptions and coverage. Never write them back as official grades.
Grade changes require a new authoritative version and audit history; update projections and flag any stale personal estimates.

### Event-driven workflow contract

Proposed command: CreatePrivateStudyBlock.
Input: personal scope, assignment reference, start/end, expected source/resource revision, idempotency key.
Server responsibilities: derive identity from trusted session; resolve authorized assignment access; validate time, conflict policy and freshness; reject invalid resource versions; bind any confirmation to exact input.
Transaction: persist private block, immutable audit reference and outbox event together.
Proposed event: personal.study_block.created.v1.
Envelope: event ID, schema version, producer identity, occurred-at, personal scope reference, aggregate ID/version, command ID, correlation ID, causation ID and restricted block reference. Sensitive references stay in scope-protected transport. Broad operational telemetry uses a separate sanitized health event, not the raw personal event. Event receipt is not permission to read the referenced object.

Dispatcher: claim an outbox lease; deliver only to authorized consumers; record delivery receipts; retry transient failures with bounded policy; quarantine permanent failures. Retention and retry limits are deployment decisions, not specified performance promises.
Consumer: authenticate producer; validate schema and scope; atomically deduplicate by consumer/event ID and update its projection; prevent older aggregate versions overwriting newer state. Consumers that detect version gaps reconcile from the authoritative aggregate.
Delivery semantics: assume at-least-once delivery. Do not claim global exactly-once processing. Downstream external writes require their own idempotent command/receipt boundary.

### Action Center propagation

A student chooses Create study block.

1. Action Center sends the command; the click itself is not an authoritative event.
2. Server resolves personal owner and authorized assignment reference.
3. Validate deadline/source revision, overlap rules and policy.
4. Show exact consequences and obtain required confirmation.
5. Commit private block + audit + outbox.
6. Master view updates planned hours from the block reference.
7. Calendar updates a private schedule projection.
8. Action Center updates its derived next-action state; creating a block does not mark the assignment submitted.
9. Notification preferences determine whether a reminder is scheduled.
10. External calendar export runs only if connected and separately authorized; it is not part of this prototype.
11. Company operations receive sanitized service health only when necessary.
12. Return a receipt: block saved; projection deliveries current/pending; no academic record changed.

Global means consistent across authorized surfaces, not broadcast to every tenant or employee.

### Workflow lifecycle and failures

Use a per-workflow state machine, not one linear path forced on every command. Shared labels may include Draft, Submitted, Validating, Waiting for information, Waiting for approval, Executing, Completed, Failed, Escalated, Cancelled and Archived. Approval states are conditional.
Private-block creation: Requested -> Validating -> Committed; each consumer separately has Pending -> Applied / Retrying / Quarantined.
A downstream calendar failure does not undo a successfully saved private block. Keep the delivery pending and reconcile.
Unknown external write outcome: inspect authoritative state using the original operation reference before retry; do not blindly repeat.
Deadline changed after planning: preserve the student's block, mark conflict if necessary and request review; never silently rearrange the student's schedule.
Membership revoked: recheck permission at reads, commands, retrieval, exports and deliveries. Stop unauthorized new use; invalidate relevant caches and subscriptions under the documented revocation procedure.

### Three-environment boundaries

Personal owns private notes, drafts, goals, effort estimates and study blocks. Institutions do not receive them by default.
Institution owns authorized educational and administrative records. Personal views may display permitted references/projections; that does not authorize indefinite copying or export.
Company owns sales, contracts, subscriptions, implementation work and sanitized service telemetry. Company identity never grants blanket academic-record access.
Personal -> institution: explicit submission, student-authorized sharing, or another reviewed lawful basis; minimize payload and record authority.
Institution -> personal: authorized read scope or explicit permitted export, honoring retention and source restrictions.
Institution -> company: contract/configuration, minimized cases and approved metrics; restricted support access is time-bounded and audited when allowed.
Company -> institution: signed scope, configuration or release instruction; cannot bypass institutional authorization.
Separate tenant/owner constraints, database RLS, storage paths/policies, search namespaces, cache keys, event subscriptions, exports, telemetry and AI retrieval. Never rely on UI filtering alone.
Personal scope is not an unrestricted null-tenant bucket. Resolve explicit ownership.
Institutional resources require explicit tenant membership and resource relationship. Keep company organization scopes distinct from institution scopes.
AI providers may process only approved data classes under the applicable policy and agreement. Until the conflicting project AI policies are reconciled, retain the stricter no-student-records external-processing rule.

## Prototype verification script

1. Open dashboard locally. Confirm Sample and Local prototype labels.
2. Sort Due ascending and descending; activate via keyboard.
3. Filter BIO 110, then Graded. Expect one matching assignment.
4. Clear filters. Sort Released score; missing scores remain last.
5. Search argument. Expect one ENG 210 assignment.
6. Create a quiz block October 8 at 15:00 for 60 minutes. Expect a linked block and planned hours increasing.
7. Enable simulated calendar failure; create a non-overlapping block before deadline. Expect saved block with Calendar projection pending.
8. Disable failure, retry. Expect calendar current; already-applied consumer duplicates ignored.
9. Choose stale Reading response. Expect scheduling blocked.
10. Try an overlapping block or one ending after its deadline. Expect rejection.
11. Export filtered CSV. Verify headers, row count and quoted values.
12. Reload. Expect session changes gone; this is intentional in-memory behavior.
    Verification status: files generated and basic fixture-reference checks performed. Browser execution, accessibility evaluation and integration tests have not been run in this environment.

## Production acceptance gates

Repository reconciliation: map proposed fields/commands to actual existing services; remove duplication.
Authorization: wrong owner, wrong institution, revoked membership and unauthorized grade tests all deny.
Isolation: no leakage through database, storage, search, caches, exports, events or AI.
Consistency: duplicate delivery, version gaps, out-of-order events, concurrent edits and stale source tests pass.
Reliability: worker crash after commit, timeout, retry, quarantine, reconciliation and recovery drills pass.
Grade correctness: ungraded, zero earned, zero possible, mixed weights, dropped scores, extra credit and changed result rules verified with institutional owners.
Time correctness: UTC/IANA timezone mapping, DST transitions and deadline updates verified.
Accessibility: keyboard interaction, focus, labels, sorting announcements, responsive layout and contrast verified.
Activation: institutional owner approves official connections/writes; privacy owner approves sharing/AI scope; operations approves runbooks and monitoring.
Readiness must be recorded separately: designed, implemented, tested, deployed, institution-authorized, production-enabled.

### Blockers and next implementation step

Blockers: live canonical schema and existing command/projection implementations not reconciled; source API authority, grading rules, timezone policy and institutional approvals not verified.
Next branch objective: map the prototype to the existing Semester shell and store; implement one private study-block command and its master-view/calendar/Action Center projections with isolated synthetic fixtures. No official grade or submission writes in that branch.
Evidence required: exact changed paths, migrations/policies if any, test commands/results, deployment evidence if applicable, risk owner and rollback/remediation plan.